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Optimization and Cost Evaluation of an Autonomous Hybrid Renewable Energy System for Electrifying Rural Areas in Nigeria

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ABSTRACT

Reliable and affordable electricity supply remains a challenge in rural communities where grid extension is technically and economically unattractive. This paper evaluates the techno-economic and environmental performance of hybrid renewable energy systems for rural electrification in Auchi, Edo State, Nigeria, with a daily load demand of 2255 kWh and a peak load of 213.53 kW. Three fully renewable configurations PV/WT/Battery, PV/Battery, and WT/Battery are analyzed under identical resource and economic conditions using HOMER Pro. Results show that the PV–WT–Battery configuration is the optimal solution, achieving a Cost of Energy (COE) of \$0.0850/kWh, a Net Present Cost (NPC) of \$904,122, and an initial capital cost of \$502,690. The optimal system consists of 1177 kW PV, two wind turbines, 1,033 batteries, and a 198 kW converter, supplying 100% renewable electricity with zero CO₂ emissions. Excess electricity is 57.5%, while unmet load is limited to 0.0693%. Comparative analysis indicates higher costs for the WT/Battery system and higher excess generation for the PV/Battery system. The results show that the hybrid PV/WT/Battery system can provide reliable, cost-effective, and zero-emission electricity for rural electrification in Nigeria. Sensitivity analysis identifies solar irradiation and nominal discount rate as the most influential parameters affecting system economics.

1 | Introduction

The global energy sector is experiencing a rapid transition from fossil-fuel-based electricity generation to renewable and sustainable energy systems. This transition is driven by concerns

about climate change, global warming, rising fuel prices, and the need to improve energy security [1]. As a result, renewable energy technologies such as solar photovoltaic (PV), wind, hydropower, biomass, and battery energy storage systems have

Abbreviations: A, swept area of wind turbine (m²); AC, alternating current; AD, autonomy days; BAT, battery energy storage system; COE, cost of energy (\$/kWh); CO₂, carbon dioxide; Cp, power coefficient; DG, diesel generator; DOD, depth of discharge (%); ES, energy storage; H, solar irradiance (W/m²); H_{STC}, reference irradiance (1000 W/m²); HOMER, hybrid optimization model for electric renewables; HRES, hybrid renewable energy system; MG, microgrid; NPC, net present cost (\$); O&M, operation and maintenance; P_{PV} , PV output power (kW); P_{rated} , rated power under standard conditions (kW); PV, photovoltaic; RES, renewable energy source; RF, renewable fraction (%); SOC, state of charge (%); T_{cell} , PV cell temperature (°C); T_{STC} , cell temperature at standard test condition (25°C); V, wind speed (m/s); V_{ci} , cut-in wind speed (m/s); V_{co} , cut-out wind speed (m/s); V_r , rated wind speed (m/s); WT, wind turbine; η_{bat} , battery efficiency; η_{inv} , inverter efficiency; β_p , power temperature coefficient (%/°C); ρ , air density (kg/m³).

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recorded significant growth worldwide [2]. By the end of 2024, the total global installed renewable energy capacity reached approximately 4448.1 GW, with solar PV contributing about 2200 GW and wind energy about 1021 GW (see Figure 1). In addition, the global cumulative installed capacity of battery energy storage systems exceeded 150 GW, corresponding to about 363 GWh [1, 3]. This rapid growth reflects a strong global shift toward clean and low-carbon energy systems.

Despite this global progress, access to reliable electricity remains a serious challenge in many developing countries, particularly in rural and remote areas. In Nigeria, a large proportion of the rural population still lacks access to stable electricity due to weak grid infrastructure, long transmission distances, high technical losses, and frequent power outages [4–6]. Grid extension to rural communities with low electricity demand is often economically unattractive. Consequently, diesel generators are commonly used to supply electricity in these areas [7, 8]. However, diesel-based power generation is associated with high operating costs, fuel transportation challenges, noise pollution, and significant carbon dioxide (CO₂) emissions, which negatively impact the environment and human health [9, 10].

Hybrid renewable energy systems (HRES), which integrate two or more renewable energy sources with energy storage systems, have emerged as a reliable and sustainable alternative for rural electrification [11]. By combining solar PV, wind turbines, and battery energy storage, HRES can improve power reliability, reduce dependence on fossil fuels, and minimize greenhouse gas emissions. The complementary nature of solar and wind resources enhances system availability, while battery storage ensures continuous power supply during periods of low renewable generation [12, 13].

Micro grid-based hybrid renewable energy systems provide additional technical and economic benefits, including reduced transmission losses, improved power quality, operational flexibility, and modular system expansion [8]. These systems can operate independently from the main grid and supply electricity directly to local loads. However, the performance and feasibility of such systems strongly depend on proper system design, optimal sizing of components, accurate load modeling, and realistic assessment of renewable energy resources [6–8].

Several software tools have been developed to support the techno-economic analysis and optimization of hybrid renewable

energy systems, including RETScreen, iHOGA, Hybrid2, and HOMER [13]. Among these tools, the Hybrid Optimization Model for Electric Renewables (HOMER) has been widely adopted due to its ability to perform hourly simulations, system optimization, and sensitivity analysis for both grid-connected and standalone systems [13–20]. HOMER allows the comparison of multiple system configurations based on economic indicators such as Net Present Cost (NPC) and Cost of Energy (COE), while satisfying technical and operational constraints [11], [21–23].

Many recent studies have applied HOMER to evaluate hybrid renewable energy systems for rural electrification, residential buildings, healthcare facilities, and critical infrastructure [13, 14, 16, 21–24]. These studies confirm that hybrid PV–wind–battery systems can significantly reduce electricity costs and CO₂ emissions compared to conventional diesel-based systems [8, 14, 25]. However, for Nigeria's South-South region, comprehensive studies that combine long-term meteorological data, reliability assessment, sensitivity analysis, and fully renewable system configurations without diesel backup remain limited.

This study presents a comprehensive techno-economic assessment of an autonomous PV–wind–battery hybrid renewable energy system for rural electrification in Auchi, Edo State, Nigeria. The novelty of this study lies in presenting the first diesel-free, fully renewable hybrid system design for Auchi community, Nigeria, demonstrating that reliable and cost-effective rural electrification can be achieved using optimized PV–wind–battery integration, and extensive sensitivity analysis to identify the most reliable and economical configuration. Unlike many previous studies, this work focuses on achieving zero-emission electricity supply with minimal unmet load, while clearly analyzing the economic sensitivity of the system to key financial and technical parameters. The remainder of the paper is organized as shown in Figure 2.

2 | Methodology and Materials

This research explores the economic feasibility of using a HRES for electricity generation in a Nigerian community. The study focuses on Auchi, a town in Edo State. Variables for optimization include solar PV, WT, and batteries, with HOMER software used to analyze the model. The technique

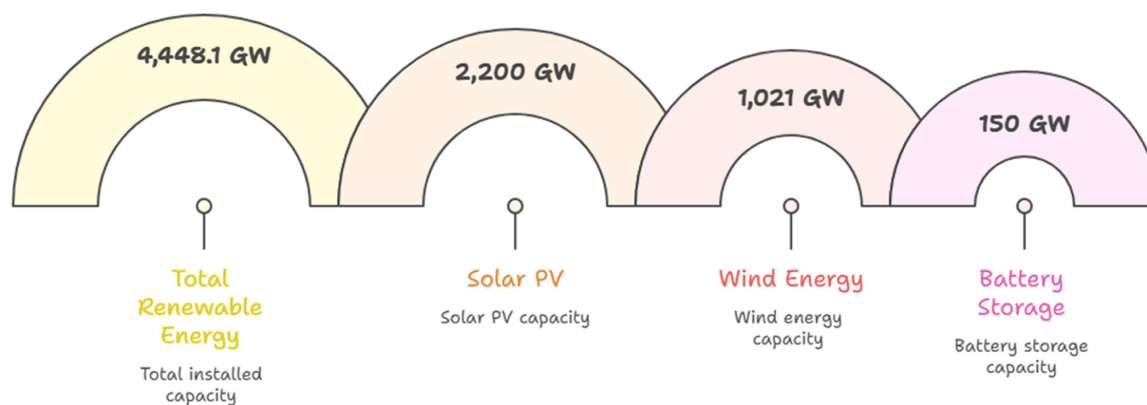


FIGURE 1 | Global renewable energy capacity by 2024 [5].



FIGURE 2 | Paper organization sequence.

enables the comparison of various configurations, assessing the technical and economic advantages of each component. The details of the study location, optimization technique, and the RES implemented are discussed below.

2.1 | Study Area Description

The study area is Auchi, located in Etsako West Local Government Area of Edo State, Nigeria. Auchi lies between latitude 7.07° N and longitude 6.27° E as shown in Figure 3. The region is characterized by a tropical climate with distinct wet and dry seasons. The average ambient temperature ranges between 23°C and 34°C throughout the year, with relatively high solar exposure due to its proximity to the equatorial region.

Electricity supply in Auchi and surrounding rural communities is unreliable due to frequent grid outages, limited transmission capacity, and high dependence on diesel generators. Grid extension to many nearby rural settlements remains economically challenging due to low load density and long feeder distances. These conditions make Auchi a suitable location for assessing the feasibility of an autonomous hybrid renewable energy system for rural electrification.

The selection of Auchi as the study location is based on three key factors: high availability of solar resources, moderate wind potential, and persistent electricity supply challenges. These characteristics make the area appropriate for evaluating a hybrid PV–wind–battery energy system as a sustainable and environmentally friendly alternative to conventional diesel-based electricity generation [4, 26].

2.2 | Solar Energy Resource Assessment

Solar energy resource assessment is a critical step in the design and optimization of PV-based HRES. In this study, solar radiation data for Auchi were obtained from the National Aeronautics and Space Administration (NASA) Surface Meteorology and Solar Energy database. The dataset covers a long-term period of 22 years, ensuring reliable representation of seasonal and annual variations in solar irradiance.

The average global horizontal irradiance in Auchi is relatively high, reflecting favorable climatic conditions for PV power generation. Monthly solar radiation values show higher

irradiance during the dry season, while slightly reduced values are observed during the rainy season due to increased cloud cover as presented in Table 1. However, the annual solar energy potential remains sufficient to support reliable PV system operation throughout the year.

The clearness index indicates moderate atmospheric transparency, which further confirms the suitability of the study area for solar energy deployment. The long-term solar data used in this study reduces uncertainty in energy yield estimation and improves the accuracy of system sizing and economic analysis. Figure 4 shows the daily irradiance and clearness index of Auchi community.

The assessed solar resource demonstrates that photovoltaic technology can serve as a primary energy source in the proposed hybrid renewable energy system. Integration with wind energy and battery storage further enhances system reliability and ensures continuous electricity supply during periods of low solar generation [11, 26].

2.3 | Wind Energy Resource Assessment

Assessment of wind energy potential is essential for determining the feasibility of integrating wind turbines into HRES. Wind speed data for Auchi, Edo State, were obtained from the NASA Surface Meteorology and Solar Energy database. The dataset spans a period of 22 years, which provides a reliable representation of long-term wind characteristics and reduces uncertainty in system design.

The wind regime in the study area is characterized by low to moderate wind speeds, which are typical of inland locations in southern Nigeria. Monthly average wind speed values indicate seasonal variation, with relatively higher wind speeds observed during the dry season and lower values during the rainy season as shown in Table 1 and Figure 5. Although the average wind speed is not suitable for large-scale wind power generation, it is adequate for small-scale wind turbines designed for low wind speed applications.

To statistically describe the wind speed distribution, the Weibull probability density function is commonly used in wind energy assessment studies [26]. The Weibull distribution allows accurate estimation of wind energy potential by accounting for wind speed variability [28]. Previous studies in similar climatic regions have demonstrated that small wind turbines can effectively complement solar PV systems by supplying power during periods of reduced solar availability, particularly at night and during cloudy conditions [27–30].

The inclusion of wind energy in the proposed hybrid system is justified by its complementary behavior with solar energy. While solar PV output is limited to daytime hours, wind energy can contribute to electricity generation during both day and night [30–32]. This complementary relationship improves system reliability, reduces battery cycling stress, and lowers the overall cost of energy.

Based on the long-term wind data and regional wind characteristics, the integration of a low-rated wind turbine into the hybrid PV/wind/battery system is considered technically feasible and beneficial for improving power supply reliability in the study area [13, 31–37].

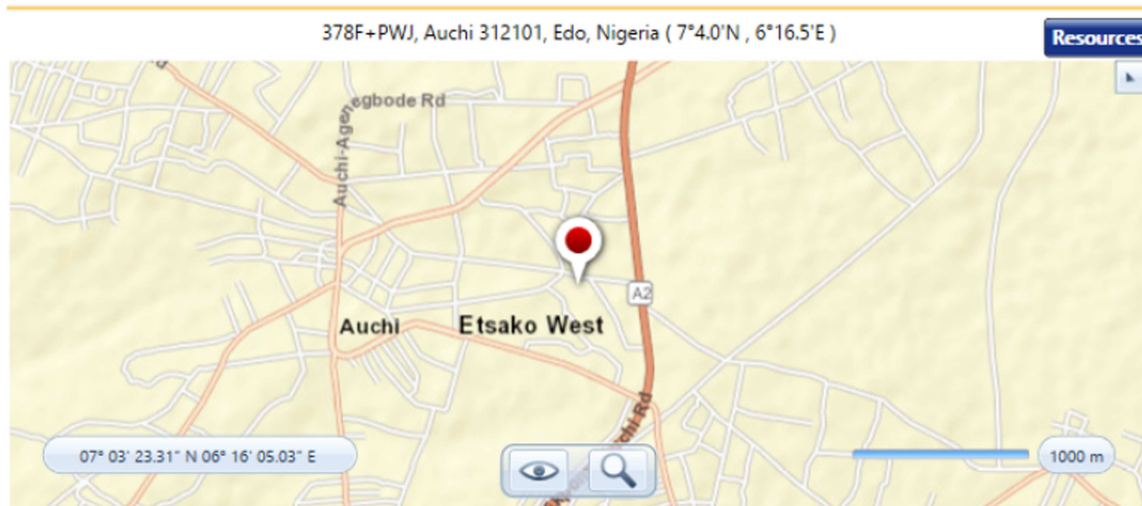


FIGURE 3 | Map of the research area.

TABLE 1 | Clearness index, average solar radiation, and wind speed profile.

Month	Clearness index	Solar radiation (kWh/m ² /day)	Wind speed (m/s)
January	0.623	5.770	3.710
February	0.594	5.840	4.030
March	0.552	5.710	4.550
April	0.518	5.420	4.790
May	0.502	5.130	4.410
June	0.469	4.700	4.500
July	0.431	4.340	4.940
August	0.401	4.130	4.930
September	0.420	4.330	4.070
October	0.484	4.800	3.420
November	0.577	5.400	3.010
December	0.619	5.590	3.310

2.4 | Load Profile and Demand Assessment

Accurate load assessment is a fundamental requirement for the proper design and optimization of HRES. In this study, the electrical load profile represents the typical electricity demand of a rural community in Auchi, Edo State. The load includes residential, small commercial, and community-based electricity usage.

The daily load profile is characterized by two peak demand periods. The first peak occurs in the morning hours due to lighting, refrigeration, and basic household activities. The second and higher peak occurs in the evening when lighting, television, and other domestic appliances are widely used. During late-night and early-morning hours, electricity demand decreases significantly as shown in Figures 6 and 7.

The annual electrical energy demand is obtained by aggregating the hourly load over 8,760 h. Seasonal variations in load

demand are relatively small compared to variations in renewable energy resources. This is because household electricity consumption in the study area is mainly driven by daily usage patterns rather than seasonal heating or cooling demands.

The load data used in this study are based on realistic assumptions derived from field observations, standard rural electrification practices, and previous studies conducted in similar socio-economic and climatic conditions. The selected load profile ensures that the proposed HRES is designed to meet actual community electricity needs without overestimation, which could lead to unnecessary capital costs.

The load profile is implemented in HOMER Pro as an hourly time series, allowing accurate simulation of system performance, energy balance, and reliability throughout the year [8, 11].

2.5 | System Configuration and Architecture

The proposed HRES consists of a PV array, a wind turbine, a battery energy storage system, a power converter, and an AC load. The system is designed to operate in standalone mode without connection to the national grid, making it suitable for rural electrification applications.

The PV array and wind turbine serve as the primary energy generation sources. Electricity generated from these sources is first used to meet the load demand. Excess energy produced during periods of high solar irradiance or strong wind conditions is stored in the battery bank through the converter. During periods of low renewable energy generation, the battery system supplies stored energy to ensure continuous power supply to the load.

The power converter performs both inversion and rectification functions. It converts DC power generated by the PV system and stored in the batteries into AC power required by the load. The converter also manages power flow between system components while maintaining voltage and frequency stability.

The hybrid configuration is selected to exploit the complementary nature of solar and wind resources. Solar PV generates

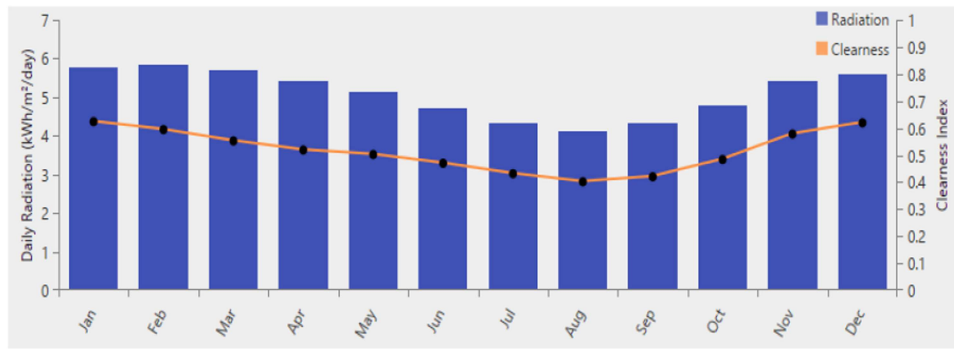


FIGURE 4 | Daily irradiance and clearness index.

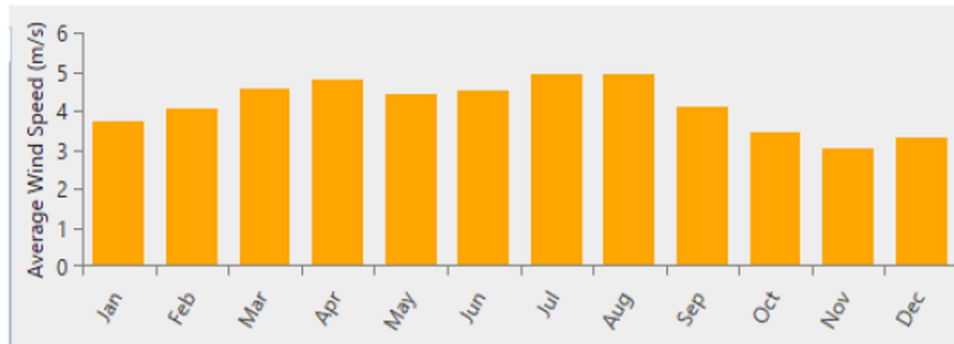


FIGURE 5 | Monthly wind speed trends.



FIGURE 6 | Hourly demand pattern.

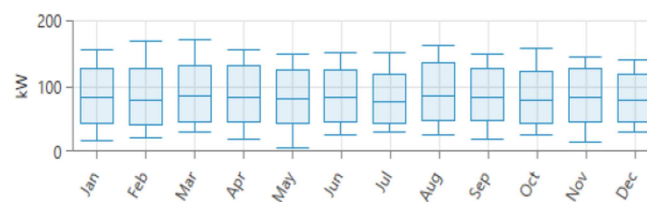


FIGURE 7 | Monthly demand pattern.

electricity primarily during daytime hours, while wind energy can contribute during both daytime and nighttime periods. Battery storage enhances system reliability by compensating for short-term fluctuations in renewable energy generation and load demand.

The system architecture is modeled and simulated using HOMER Pro, which evaluates energy balance, component interactions, and system performance on an hourly basis. This configuration enables optimal sizing of system components

while ensuring technical feasibility, economic efficiency, and reliable electricity supply [38–40].

2.6 | Technical Specifications of System Components

Proper selection of system components is essential for achieving reliable operation and cost-effective performance of the hybrid renewable energy system. The technical specifications of the main components used in this study are summarized and justified in the following subsections.

2.6.1 | Photovoltaic Module

The PV module used in this study is a monocrystalline silicon module, chosen due to its high conversion efficiency and better temperature performance compared to polycrystalline or thin-film modules [41, 42]. Monocrystalline modules are widely used in both urban and off-grid applications, particularly where space is limited and higher efficiency is needed. Their superior performance under partial shading and higher irradiance levels makes them suitable for hybrid PV systems, as they maintain relatively higher electrical output even when the surface temperature rises slightly [43]. The technical specification of the PV module is highlighted in Table 2 with a derating factor of 80% during a lifespan of 25 years.

The selected module has the following key characteristics:

The power temperature coefficient (β) indicates the relative change in power output per degree Celsius increase in cell temperature. A negative value ($-0.35\%/^{\circ}\text{C}$) shows that as the cell temperature increases, the module output decreases.

TABLE 2 | Technical specification of the PV panel.

Parameter	Value	Unit
Rated power (P_{STC})	320	W
Open circuit voltage (V_{oc})	40.2	V
Short-circuit current (I_{oc})	9.68	A
Maximum power Voltage (V_{mp})	33.1	V
Maximum power current (I_{mp})	9.68	A
Power temperature coefficient (β)	-0.35	%/°C
Nominal operating cell temperature (NOCT)	45 $\pm$ 2	°C

By including this coefficient in the model, the study accounts for real operating conditions and allows accurate prediction of electrical output in the hybrid PV system.

2.6.2 | Wind Turbine

A small-scale horizontal-axis wind turbine is selected for the hybrid system. The turbine is designed for low wind speed applications, with a low cut-in wind speed suitable for the moderate wind conditions observed in the study area.

The wind turbine is defined by its rated power, cut-in speed, rated speed, cut-out speed, and hub height. The selection of a small wind turbine complements the solar PV system by providing additional power during night-time and low solar radiation periods. This improves system reliability and reduces battery stress [16, 30]. Table 3 presents the wind turbine technical specifications used in this research.

2.6.3 | Battery Energy Storage System

The battery energy storage system is based on deep-cycle batteries suitable for renewable energy applications. These batteries are designed to withstand frequent charge–discharge cycles and provide reliable energy storage over long periods.

Key battery parameters include nominal voltage, capacity, depth of discharge, round-trip efficiency, and lifetime throughput. The allowable depth of discharge is limited to extend battery lifespan and ensure reliable system operation. Temperature effects on battery performance are considered through a correction factor applied during capacity estimation [41–44]. Table 4 presents the cost specifications of the hybrid system components.

2.6.4 | Power Converter

The power converter is responsible for converting DC power from the PV array and battery bank into AC power for the load. The converter is defined by its rated capacity, efficiency, and bidirectional power flow capability.

The converter capacity is selected to handle peak load demand and ensure stable operation under varying generation and load conditions. High-efficiency converters are preferred to minimize energy losses and improve overall system performance [45–47].

TABLE 3 | Technical specifications for the wind turbine.

Parameter	Value	Unit
Rated capacity	20	kW
Vci	2.5	m/s
Vco	20	m/s
Vr	5	m/s
Capacity factor	20.8	%

TABLE 4 | System component specifications.

	Parameter	Value	Unit
PV panel	Ground reflection	20	%
	Capital cost/kW	200	\$
	Replacement cost/kW	200	\$
	Operational & maintenance cost	10	\$
	Lifetime	12	Years
WT	Capital cost/kW	7000	\$
	Replacement cost/kW	7000	\$
	Operational & maintenance cost	35	\$
	Lifetime	20	Years
Converter	Capital cost/kW	350	\$
	Replacement cost/kW	350	\$
	Operational & maintenance cost	10	\$
	lifetime	10	Years
	Inverter's efficiency	96	%
Battery	Rectifier's efficiency	94	%
	Capital cost	200	\$
	Replacement cost	200	\$
	Operational & maintenance cost	10	\$
	Efficiency	95	%
	Lifetime	12	Years
	Minimum SOC	40	%

2.6.5 | Assumptions, Limitations, and Uncertainties

The techno-economic assessment carried out in this study is based on several assumptions that are necessary for system modeling and optimization. These assumptions are made to ensure realistic system behavior while maintaining computational efficiency.

The following assumptions are adopted:

- Renewable energy resources (solar irradiance and wind speed) are represented using long-term averaged meteorological data obtained from NASA.

- Component performance degradation is implicitly handled within the HOMER Pro simulation framework.
- All system components operate within their rated technical limits.
- Battery operation is restricted by predefined minimum and maximum state of charge limits.

Despite careful modeling, certain limitations and uncertainties remain. Renewable energy resources are inherently variable, and actual site conditions may differ slightly from historical averages. Load demand may also increase over time due to population growth and increased electricity usage. Economic parameters such as discount rate, component costs, and operation and maintenance costs are subject to market fluctuations. These uncertainties are addressed through sensitivity analysis, which evaluates the impact of key parameters on system performance and cost.

2.7 | System Modeling and Optimization

2.7.1 | Solar Photovoltaic (PV) System Modeling

The electrical power output of the PV system is modeled based on the rated power under standard test conditions and adjusted for real operating conditions. The PV output power is expressed as in [29]

$$P_{pv} = P_{STC} [1 + \beta(T_c - T_{STC})] \frac{H}{H_{STC}} \quad (1)$$

where T_c is the cell surface temperature under operating conditions, T_{STC} is the cell temperature at standard test conditions (25°C), β is the power temperature coefficient (%/°C), H is the solar irradiance incident on the PV surface, and H_{STC} is the irradiance under STC (1000 W/m²). To estimate T_c practically, an empirical relation proposed by [29] was adopted:

$$T_c = T_{\infty} + 0.078 \cdot H_t \quad (2)$$

where T_{∞} is the ambient temperature in °C and H_t is the solar irradiance in W/m². This approach captures the influence of both ambient temperature and irradiance on the PV cell, allowing a more realistic prediction of electrical performance.

2.8 | Wind Turbine Modeling

The electrical power generated by the wind turbine depends on air density, swept area, power coefficient, and wind speed. The mechanical power captured from wind is given as in [12]

$$P_m = \frac{\rho A C_p V^3}{2} \quad (3)$$

where C_p represents the power coefficient, A is the swept area, ρ denotes the air density, and V is the wind velocity. The power captured at a specific wind velocity and time t is computed using Equation (3) [30, 31]:

$$P_w(t) = \begin{cases} 0, & \text{if } V(t) \geq V_{co} \text{ or } V(t) \leq V_{ci} \\ P_{rated} \times \left(\frac{V^3(t) - V_{ci}^3}{V_r^3 - V_{ci}^3} \right) & \text{if } V_{ci} < V(t) \text{ and } V(t) < V_r \\ P_{rated} & V_r < V(t) \text{ and } V(t) < V_{co} \end{cases} \quad (4)$$

where P_{rated} is the rated wind turbine power (Kw), V_{ci} (m/s) is the cut in speed, V_{co} (m/s) is the cut out speed, V (m/s) is the location's hourly wind speed, and V_r (m/s) is the rated turbine wind speed.

2.9 | Battery Energy Storage Modeling

The battery energy storage system is used to store excess renewable energy and supply power during periods of low generation. The state of charge of the battery at time t is calculated as in [45]

$$SoC(t) = SoC(t-1) \cdot (1 - \sigma) + \frac{P_{in}(t) - P_{load}(t)}{\eta \times P_{max}} \quad (5)$$

where $SoC(t-1)$ is the state-of-charge at the previous time step, σ is self-discharge rate of the battery, $P_{in}(t)$ is the power supplied to the battery at time t , $P_{load}(t)$ is the power drawn from the battery at time t , η is the battery efficiency, while P_{max} is the maximum battery capacity.

The battery capacity includes a temperature correction factor to account for performance variations due to ambient and operational temperatures. The correction factor CF can be applied as [48]

$$CF = 0.0004 T^2 - 0.0246 T + 1.3961 \quad (6)$$

where T is the ambient temperature in °C. The effective maximum usable battery capacity then becomes:

$$P_{max} = DOD \times CF \times P_{rated} \quad (7)$$

where DOD is the depth-of-discharge and P_{rated} is the nominal rated capacity of the battery.

This formulation ensures reliable battery operation under varying temperature conditions [5].

2.10 | Power System Management Approach

The operation strategy is designed so that, under normal conditions, the energy generated by the RESs meets the load demand. When generation exceeds demand, the surplus energy charges the batteries up to 100%. Any further excess energy is sent to a dump load. During apex demand periods, if the RES cannot fully supply the load.

2.11 | Optimization Objective Function and Constraints

The optimization process aims to identify the most economical and reliable HRES configuration that satisfies the electrical demand of Auchic community.

2.12 | Objective Function

The optimization problem is formulated to minimize the total net present cost (NPC) of the hybrid renewable energy system. The cost of energy (COE) is evaluated as a derived economic metric. System reliability is ensured by constraining unmet load and capacity shortage within acceptable limits, while environmental benefits such as CO₂ emission reduction are assessed as post-optimization performance indicators.

The optimal system configuration is determined by minimizing the total NPC of the system over the project lifetime.

Mathematically, this is expressed as

min NPC

With NPC defined as the life-cycle cost of the hybrid system:

$$NPC = C_{\text{cap}} + \sum_{y=1}^N \frac{C_{\text{O\&M}}(y) + C_{\text{rep}}(y) + C_{\text{fuel}}(y)}{(1+r)^y} - \frac{C_{\text{salv}}}{(1+r)^N} \quad (7)$$

where C_{cap} is the initial capital cost, $C_{\text{O\&M}}$ is operation and maintenance cost, C_{rep} is the replacement cost, C_{fuel} is fuel cost (zero for this system), C_{salv} is salvage value, r is discount rate while N denote the project lifetime (years).

The cost of energy (COE) is used to evaluate the economic feasibility of the system as given [49] which directly depends on the NPC of the system:

$$\text{COE} = \frac{\text{NPC} \times \text{CRF}(r, N)}{E_{\text{served}}} \quad (8)$$

where

$$\text{CRF}(r, N) = \frac{r(1+r)^N}{(1+r)^N - 1} \quad (9)$$

HOMER Pro evaluates all feasible system configurations and selects the optimal solution based on the lowest NPC while meeting technical and operational requirements [33].

2.12.1 | Constraints

The optimization is subject to the following constraints:

2.12.1.1 | Power Balance Constraint

Total power generated by PV, wind turbines, and batteries must meet the load demand at every time step with unmet load and surplus explicitly allowed as in [50, 51]

$$P_{\text{PV}}(t) + P_{\text{WT}}(t) + P_{\text{bat}}^{\text{dis}}(t) + P_{\text{unmet}}(t) = P_L(t) + P_{\text{bat}}^{\text{ch}} + P_{\text{dump}}(t) \quad (10)$$

With non-negativity:

$$P_{\text{unmet}}(t) \geq 0, \quad P_{\text{dump}}(t) \geq 0 \quad (11)$$

2.12.1.2 | Battery State of Charge Constraint As in [51]

$$\text{SOC}_{\text{min}} \leq \text{SOC}(t) \leq \text{SOC}_{\text{max}} \quad (12)$$

While the charging/discharging limits are as follows:

$$0 \leq P_{\text{bat}}^{\text{ch}}(t) \leq P_{\text{bat}}^{\text{ch, max}}, \quad 0 \leq P_{\text{bat}}^{\text{dis}}(t) \leq P_{\text{bat}}^{\text{dis, max}}(t) \quad (13)$$

2.12.1.3 | Renewable Resource Availability Constraint

Power generation depends on available solar irradiance and wind speed. This is typically enforced by upper bounding generation by the resource driven available power as in [52]

$$0 \leq P_{\text{PV}}(t) \leq P_{\text{PV}}(G(t), T(t)) \quad (14)$$

$$0 \leq P_{\text{WT}}(t) \leq P_{\text{WT}}(V(t)) \quad (15)$$

where $P_{\text{WT}}(t)$ and $P_{\text{PV}}(t)$ are the wind and PV models as given in Equations (1) and (4).

2.12.1.4 | Converter Capacity Constraint

The inverter and rectifier capacities must not be exceeded.

For $DC \rightarrow AC$ inverter and $AC \rightarrow DC$ rectifier as in [53]

$$0 \leq P_{\text{int}}(t) \leq P_{\text{int}}^{\text{rated}} \quad (16)$$

$$0 \leq P_{\text{rec}}(t) \leq P_{\text{rec}}^{\text{rated}} \quad (17)$$

2.12.1.5 | Reliability Constraint

Unmet load and capacity shortage must remain within acceptable limits. A common energy-based reliability constraint is as in [53]

$$\frac{\sum_{t=1}^T E_{\text{unmet}}(t)}{\sum_{t=1}^T E_L(t)} \leq \epsilon \quad (18)$$

where $E_{\text{unmet}}(t) = P_{\text{unmet}}(t)\Delta t$, $E_L(t) = P_L(t)\Delta t$, and ϵ is allowed unmet load fraction which should be $\leq 1\%$. And the capacity shortage is given as

$$\max_{t \in p} \left(\frac{P_{\text{unmet}}(t)}{P_L(t)} \right) \leq \epsilon \quad (19)$$

These constraints ensure stable and continuous operation of the hybrid energy system.

2.12.1.6 | Justification for Using Homer Pro

HOMER Pro is used in this study due to its proven capability in modelling, optimizing, and analyzing hybrid renewable energy systems. The software performs hourly simulations over an annual cycle, accounting for variations in load demand and renewable resource availability. HOMER integrates technical performance, economic evaluation, and sensitivity analysis within a single framework as shown in Figure 8, making it suitable for off-grid and rural electrification studies [54–56].

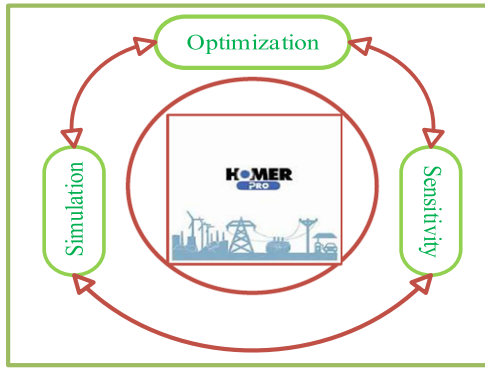


FIGURE 8 | Functions of Homer pro.

HOMER employs validated component models and optimization routines to identify optimal system configurations based on economic indicators such as NPC and COE. While HOMER relies on established solar radiation transposition and component performance models, uncertainties associated with these models are acknowledged and addressed through sensitivity analysis and conservative design assumptions.

3 | Environmental and CO₂ Emission Analysis

Environmental impact assessment is an important aspect of hybrid renewable energy system planning, particularly in regions where electricity generation is traditionally dominated by fossil fuel sources. In Nigeria, diesel generators are widely used to compensate for unreliable grid supply, resulting in high fuel consumption and significant carbon dioxide (CO₂) emissions. The proposed hybrid PV-wind-battery system offers a clean and sustainable alternative with minimal environmental impact.

3.1 | CO₂ Emission Reduction

The hybrid renewable energy system considered in this study operates without diesel backup and therefore does not produce direct CO₂ emissions during operation. This represents a major environmental advantage compared to conventional diesel-based power systems.

To quantify the environmental benefit, CO₂ emissions avoided by the hybrid system are estimated by comparing the annual electrical energy supplied by the system with an equivalent diesel-based electricity generation scenario. The avoided CO₂ emissions are calculated as

$$\text{CO}_2 = \text{EF}_{\text{CO}_2} \times E_t \quad (20)$$

where EF_{CO_2} is the emission factor of diesel-based electricity generation (kg CO₂/kWh) and E_t is the annual electrical energy supplied by the hybrid system (kWh) [34].

The significant reduction in CO₂ emissions contributes to climate change mitigation and supports national and global targets for reducing greenhouse gas emissions.

3.1.1 | Environmental Cost and Economic Impact

The cost of environmental damage caused by CO₂ emissions can be incorporated into economic analysis to provide a more comprehensive evaluation of energy systems. The environmental damage cost associated with CO₂ emissions is calculated as

$$C_{\text{CO}_2} = \text{EF}_{\text{CO}_2} \times E_t \times \phi_{\text{CO}_2} \quad (21)$$

where ϕ_{CO_2} represents the social cost of carbon, which is considered as 70 USD per ton of CO₂ [35].

To account for the environmental impact of electricity generation, the economic analysis of the hybrid PV system has been extended to include the cost of CO₂ emissions. This allows evaluation not only of financial feasibility but also of environmental benefits. The Levelized Cost of Energy (LCOE) and Payback Time of Money (PBTM) were adjusted to incorporate the social cost of carbon using the following relations:

$$\text{LCOE} = \frac{\left(\frac{r(1+r)^n}{(1+r)^n - 1} \right) \times C + C_{\text{O\&M}} - C_{\text{CO}_2}}{E_t} \quad (22)$$

$$\text{PBTM} = \frac{C}{\text{Income}} \quad (23)$$

where C is the total capital cost, $C_{\text{O\&M}}$ is the operation and maintenance cost, C_{CO_2} is the CO₂ damage cost, E_t is the total energy output over the system lifetime, and r is the discount rate. The CO₂ damage cost is computed as:

$$C_{\text{CO}_2} = \text{EF} \times E_t \times \phi \quad (24)$$

where EF is the CO₂ emission factor of the conventional power generation system (kg CO₂/kWh) and ϕ is the social cost of carbon (\$/ton CO₂, typically \$70/ton) [30].

Including the environmental damage cost reduces the LCOE and improves the payback time, highlighting the financial and ecological advantages of replacing conventional electricity with renewable PV energy. Studies in similar contexts, such as Libya, have shown that this approach enhances the competitiveness of renewable energy in the local market and supports sustainable energy planning [57].

3.1.2 | Role of Hybrid Renewable Energy Systems in Clean Energy Transition

The proposed hybrid PV/wind/Battery system supports the transition toward cleaner energy resources by reducing dependence on fossil fuels and minimizing greenhouse gas emissions. The system contributes to improved air quality, reduced noise pollution, and enhanced energy sustainability in rural communities.

The integration of renewable energy and energy storage technologies aligns with global efforts to promote low-carbon energy systems and supports long-term environmental and socio-economic development goals [8, 35, 49].

4 | Results and Discussion

4.1 | PV/WT/Battery Configuration

The PV/WT/Battery HRES consists of 1177 kW of solar PV capacity, two wind turbines, 1033 battery units, and a 198 kW power converter. The system achieves a COE of \$0.0850/kWh and an NPC of \$904,122. The initial capital investment required is \$502,690.

Cost distribution analysis shows that the PV system contributes 42.87% of the total NPC, while the battery system and power converter account for 34.74% and 20.52%, respectively. The wind turbine contributes only 1.86% of the NPC, reflecting its supportive role in the hybrid system.

The system supplies 100% of the electricity from renewable sources, resulting in zero operational emissions. Excess electricity generation is 57.5%, corresponding to 1,198,863 kWh/year, while unmet demand remains very low at 0.0693% (570 kWh/year). These results confirm high system reliability and effective energy balance.

4.2 | PV/Battery Configuration

The PV/Battery configuration includes 1238 kW of PV capacity, 1015 battery units, and a 203 kW power converter. The system records a COE of \$0.0852/kWh and an NPC of \$906,000, with an initial capital cost of \$501,196.

The PV system represents 44.98% of the NPC, while the battery and converter contribute 34.06% and 20.96%, respectively. Similar to the PV/WT/Battery configuration, this system supplies 100% renewable electricity with zero emissions.

However, excess electricity generation is slightly higher at 59.1% (1,279,410 kWh/year), and unmet demand increases marginally to 0.0747% (615 kWh/year). These values indicate a slightly lower energy utilization efficiency compared to the hybrid system that includes wind energy.

4.3 | WT/Battery Configuration

The WT/Battery configuration consists of 236 wind turbine generators, 3109 battery units, and a 647 kW power converter. This system exhibits a significantly higher COE of \$0.333/kWh and a total NPC of \$3,540,000, with an initial capital cost of \$2,440,000.

The wind turbine system contributes 56.18% of the initial cost, while the battery and converter account for 26.70% and 17.11%, respectively. Although the system supplies 100% renewable electricity with zero emissions, excess electricity generation reaches 68.9% (1,921,707 kWh/year), indicating substantial overproduction.

Despite achieving the lowest unmet demand of 0.0314% (258 kWh/year), the very high capital cost and energy cost make this configuration economically unattractive for rural electrification.

4.4 | Comparison of Hybrid System Configurations

The comparison results clearly show significant differences in economic and technical performance among the three hybrids renewable energy system configurations, despite all systems achieving 100% renewable energy supply and zero operational emissions as shown in Table 5.

The PV/WT/Battery configuration demonstrates the lowest NPC (\$904,122) and the lowest COE (\$0.0850/kWh) among the three options. The inclusion of wind energy, even at a modest capacity, improves system flexibility and reduces excessive dependence on battery storage. This results in better energy utilization and lower excess electricity compared to the PV–Battery configuration.

The PV/Battery configuration shows economic performance close to that of the PV/WT/Battery system. However, the absence of wind energy leads to higher excess electricity generation and slightly higher unmet load. This indicates reduced system complementarity and increased battery cycling, which may affect long-term battery lifespan and system robustness.

TABLE 5 | Comparison of optimal hybrid renewable energy system configurations.

Parameter	PV/WT/battery	PV/battery	WT/battery
PV capacity (kW)	1,177	1,238	---
Number of wind turbines	2	---	236
Number of batteries	1,033	1,015	3,109
Converter capacity (kW)	198	203	647
Initial capital cost (USD)	502,690	501,196	2,440,000
Net present cost, NPC (USD)	904,122	906,000	3,540,000
Cost of energy, COE (USD/kWh)	0.0850	0.0852	0.333
Renewable fraction (%)	100	100	100
Excess electricity (%)	57.5	59.1	68.9
Excess electricity (kWh/year)	1,198,863	1,279,410	1,921,707
Unmet load (%)	0.0693	0.0747	0.0314
Unmet load (kWh/year)	570	615	258
CO ₂ emissions	0	0	0

The WT/Battery configuration achieves the lowest unmet load but at a very high economic cost. The system records an NPC of \$3.54 million and a COE of \$0.333/kWh, which is nearly four times higher than the other configurations. The extremely high excess electricity generation indicates significant overdesign, making this configuration economically impractical for rural electrification.

Overall, the comparison confirms that hybridization of solar PV, wind energy, and battery storage provides the most balanced solution, offering low energy cost, high reliability, and reduced system oversizing. The PV/WT/Battery configuration therefore represents the optimal system for the study area when economic efficiency, technical reliability, and environmental sustainability are jointly considered.

4.5 | Sensitivity Analysis

Sensitivity analysis is performed to evaluate the influence of key technical and economic parameters on the system performance and economic indicators. In this study, the

sensitivity assessment focuses on variations in solar irradiation (represented by the CAT-TimeShift derating factor), wind turbine hub height, and nominal discount rate. The impact of these parameters is examined in terms of COE, NPC, initial capital cost, and O&M cost.

The selected parameters are varied within realistic ranges to reflect possible changes in climatic conditions and economic scenarios. Table 6 presents the detailed results of the sensitivity analysis.

The sensitivity analysis results show that nominal discount rate has the most significant impact on both COE and NPC. An increase in the discount rate leads to higher COE and NPC values, indicating strong economic sensitivity to financial conditions. Conversely, lower discount rates substantially improve system feasibility and reduce overall energy cost.

Solar irradiation, represented by the CAT-Time Shift derating factor, also exerts a strong influence on system performance. Higher solar availability leads to lower COE and NPC due to increased PV energy production and reduced reliance on battery storage. This confirms the dominant role of solar energy in

TABLE 6 | Sensitivity analysis results for economic indicators.

Derating factor (%)	Hub height (m)	Nominal discount rate (%)	COE (\$/kWh)	NPC (\$)	Initial cost (\$)	O&M cost (\$)
100	13	8	0.0856	906,000	501,196	31,426
100	24	8	0.0852	904,123	502,690	31,313
100	50	8	0.0850	1,030,000	585,666	31,053
80	13	8	0.0973	996,563	550,491	34,756
80	24	8	0.0937	996,563	550,491	34,506
80	50	8	0.0937	948,960	524,083	34,506
90	13	8	0.0893	963,621	538,278	32,866
90	24	8	0.0906	988,941	535,005	32,902
90	50	8	0.0883	840,193	504,245	31,246
100	13	10	0.0944	835,792	501,196	31,051
100	24	10	0.0939	833,028	497,390	30,926
100	50	10	0.0936	890,608	507,329	31,022
80	13	10	0.1000	890,608	507,329	35,425
80	24	10	0.1000	868,394	507,329	35,425
80	50	10	0.1000	890,608	507,329	35,425
90	13	10	0.0976	890,241	517,942	32,391
90	24	10	0.1000	890,241	538,278	32,531
90	50	10	0.0981	873,226	524,091	32,269
100	10	12	0.1040	786,044	504,245	30,578
100	100	12	0.1030	781,700	501,196	30,438
100	50	12	0.1030	780,036	502,297	30,138
90	10	12	0.1090	828,104	507,329	34,808
90	100	12	0.1090	828,104	507,329	34,808
90	50	12	0.1090	828,104	507,329	34,808
80	10	12	0.1070	811,786	517,942	31,885
80	100	12	0.1100	833,763	538,278	32,063
80	50	12	0.1080	816,838	524,091	31,766

the proposed hybrid system, which is consistent with the high solar potential of the study area.

Wind-related parameters, represented by hub height variation, show a moderate impact on system economics. Increasing hub height slightly improves energy capture and reduces system cost, but the effect remains less pronounced compared to solar irradiation and discount rate variations. This outcome reflects the moderate wind resource available at the study location.

Overall, the sensitivity analysis confirms that the proposed HRES is economically robust, particularly under favorable financial conditions and high solar resource availability. The results further support the selection of the PV/WT/Battery configuration as the optimal system for the study area.

5 | Conclusion and Recommendation

This study presents a comprehensive techno-economic assessment of HRES for rural electrification in Auchi, Edo State, Nigeria. Three system configurations, namely PV/WT/Battery, PV-Battery, and WT-Battery, are analyzed and compared using HOMER Pro based on economic performance, reliability, and environmental impact.

The results show that the PV/WT/Battery configuration provides the most balanced and cost-effective solution. This system consists of 1177 kW of PV capacity, two wind turbines, 1033 battery units, and a 198 kW power converter. The system achieves a low COE of \$0.0850/kWh and a \$904,122, while supplying electricity entirely from renewable sources with zero operational CO₂ emissions. The system also demonstrates high reliability, with minimal unmet load and acceptable levels of excess electricity generation.

Comparison with alternative configurations confirms that the PV/Battery system exhibits slightly higher energy cost and excess electricity, while the WT/Battery system suffers from extremely high capital cost and poor economic feasibility despite high reliability. These findings highlight the importance of hybridization in improving system performance and reducing overall cost.

Sensitivity analysis indicates that the system economics are strongly influenced by the nominal discount rate and solar irradiation levels. Lower discount rates and higher solar availability significantly reduce COE and NPC, while wind-related parameters have a moderate impact. This confirms the robustness of the proposed system under varying technical and economic conditions.

Although HOMER Pro provides a robust framework for techno-economic optimization, future research may extend this work by validating the obtained optimal configuration using advanced meta-heuristic optimization techniques, such as Particle Swarm Optimization, Mayfly Algorithm, Harmony Search, Flower Pollination Algorithm, or other emerging bio-inspired algorithms. Such approaches can further enhance solution robustness and allow multi-objective optimization considering cost, reliability, and environmental impact simultaneously.

Overall, the results confirm that hybrid PV/WT/Battery systems represent a reliable, economically viable, and environmentally sustainable solution for rural electrification in Nigeria. The

findings provide useful guidance for policymakers, energy planners, and rural electrification programs.

Author Contributions

Ibrahim Seidu: conceptualization, data curation, formal analysis, investigation, methodology, visualization, writing – original draft. **Sani Salisu:** conceptualization, methodology, project administration, resources, software, supervision, validation, writing – review and editing. **Abdullahi Abubakar Mas'ud:** conceptualization, methodology, project administration, resources, software, supervision, validation, writing – review and editing. **Umar Musa:** conceptualization, methodology, project administration, resources, software, supervision, validation, writing – review and editing. **Ibrahim Abdullahi Shehu:** writing – review and editing. **Adamu Abubakar Sa'idu:** writing – review and editing. **Firdaus Muhammad-Sukki:** methodology, project administration, visualization, writing – review and editing. **Mohd Khairul Amri Kamarudin:** writing – review and editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sets generated during the current study are available from the corresponding authors on reasonable request.

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